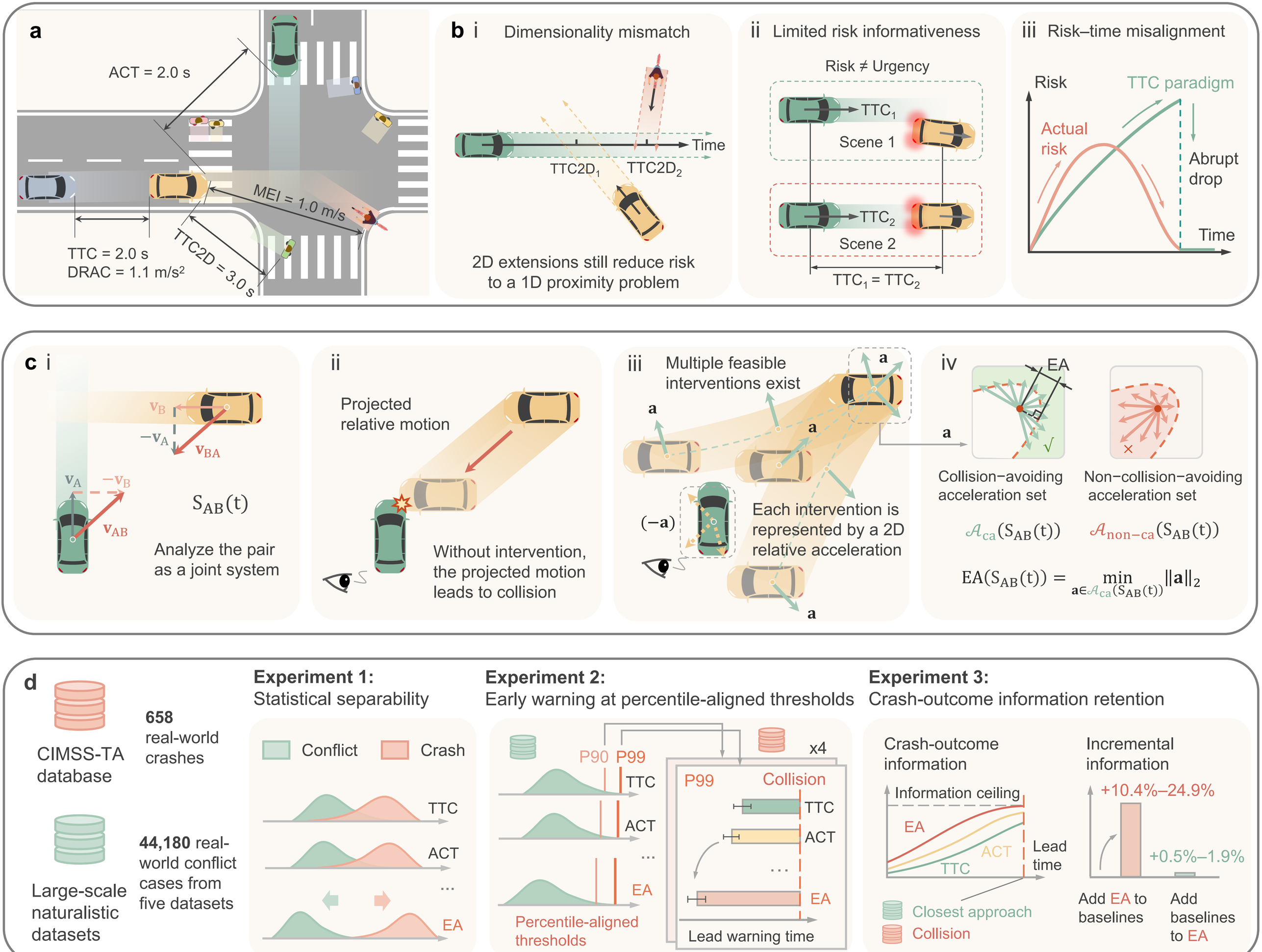

a
ACT = 2.0 s
MEI = 1.0 m/s
TTC = 2.0 s
DRAC = 1.1 m/s²
TTC2D = 3.0 s
b i Dimensionality mismatch
Time
TTC2D1
TTC2D2
2D extensions still reduce risk
to a 1D proximity problem
ii Limited risk informativeness
Risk ≠ Urgency
TTC1
Scene 1
TTC2
Scene 2
TTC1 = TTC2
iii Risk–time misalignment
Risk
TTC paradigm
Actual
risk
Abrupt
drop
Time
c i
vB
−vA
vBA
vA
−vB
vAB
SAB(t)
Analyze the pair
as a joint system
ii
Projected
relative motion
Without intervention,
the projected motion
leads to collision
iii Multiple feasible
interventions exist
a
(−a)
Each intervention is
represented by a 2D
relative acceleration
iv
EA
Collision−avoiding
acceleration set
Non−collision−avoiding
acceleration set
$\mathcal{A}_{ca}(S_{AB}(t))$
$\mathcal{A}_{non-ca}(S_{AB}(t))$
$EA(S_{AB}(t)) = \min_{a \in \mathcal{A}_{ca}(S_{AB}(t))} \|a\|_2$
d
CIMSS-TA
database
658
real-world
crashes
Large-scale
naturalistic
datasets
44,180 real-
world conflict
cases from
five datasets
Experiment 1:
Statistical separability
Conflict
Crash
TTC
ACT
...
EA
Experiment 2:
Early warning at percentile-aligned thresholds
P90
P99
TTC
ACT
...
EA
Percentile-aligned
thresholds
x4
P99
Collision
Lead warning time
Experiment 3:
Crash-outcome information retention
Crash-outcome
information
Information ceiling
EA
ACT
TTC
Lead
time
Closest approach
Collision
Incremental
information
+10.4%–24.9%
+0.5%–1.9%
Add EA to
baselines
Add
baselines
to EA